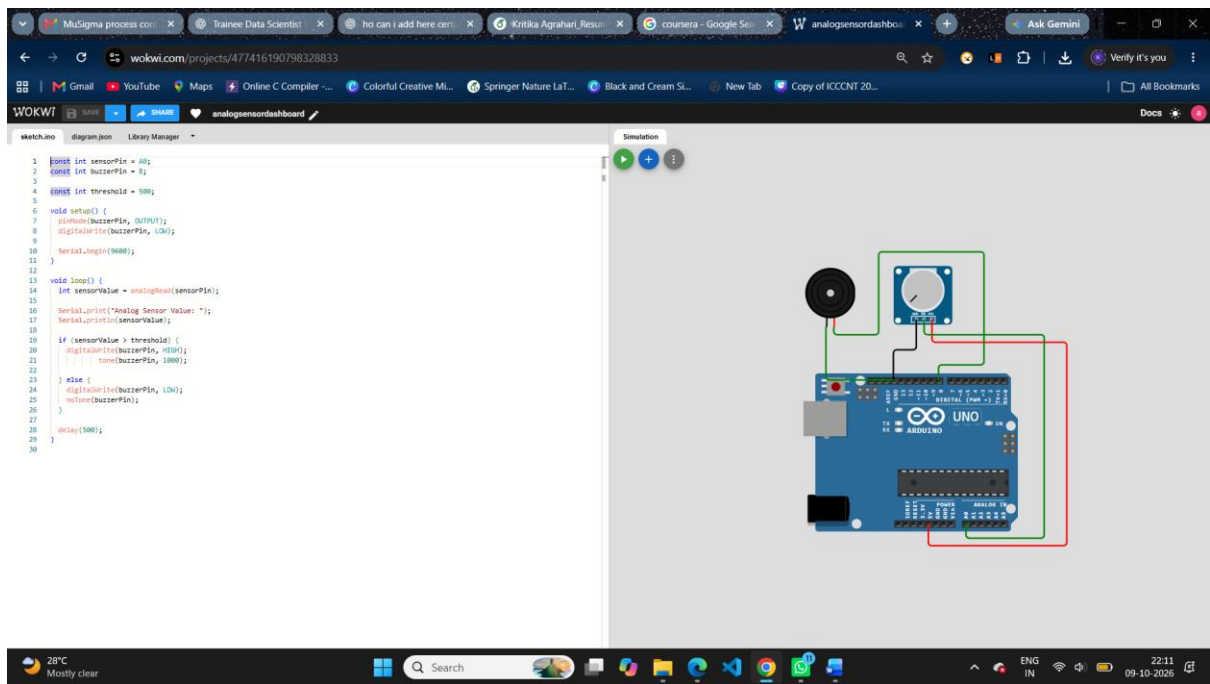
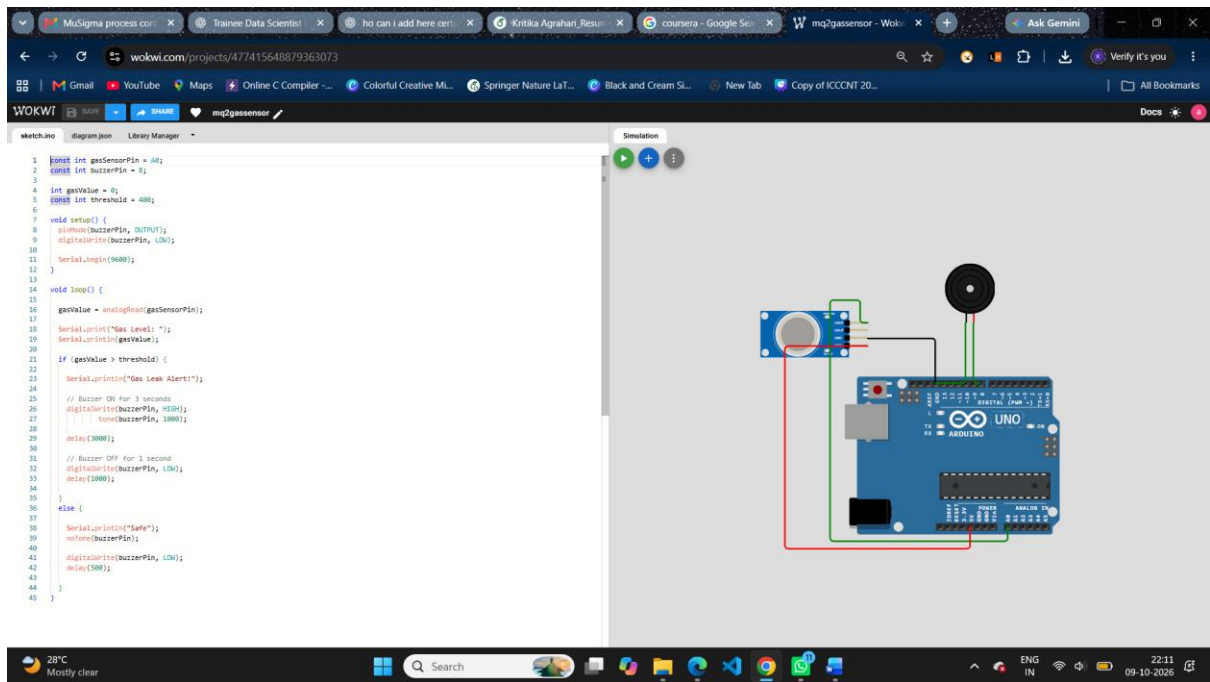
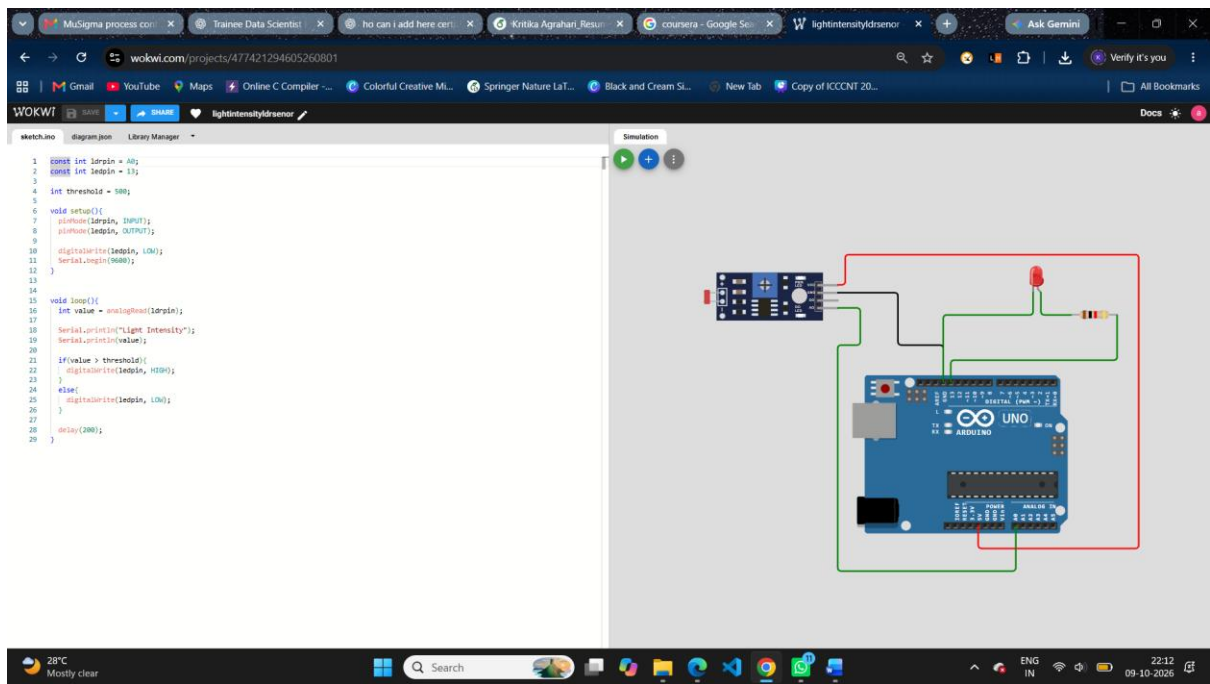
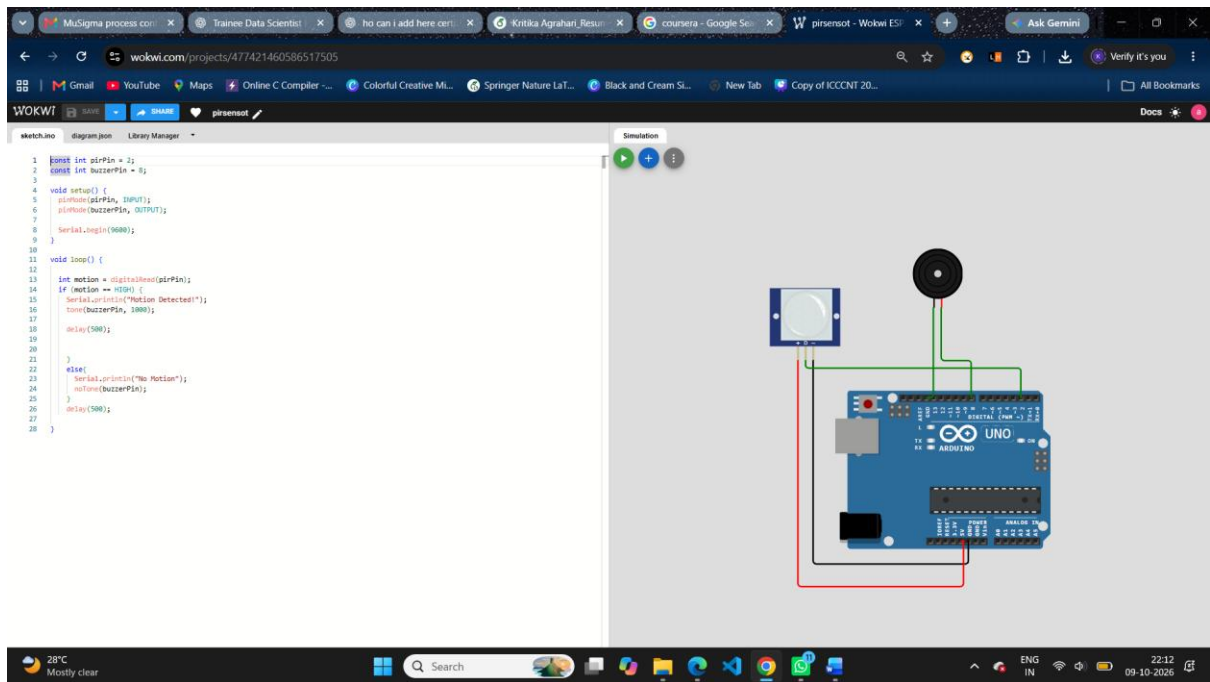
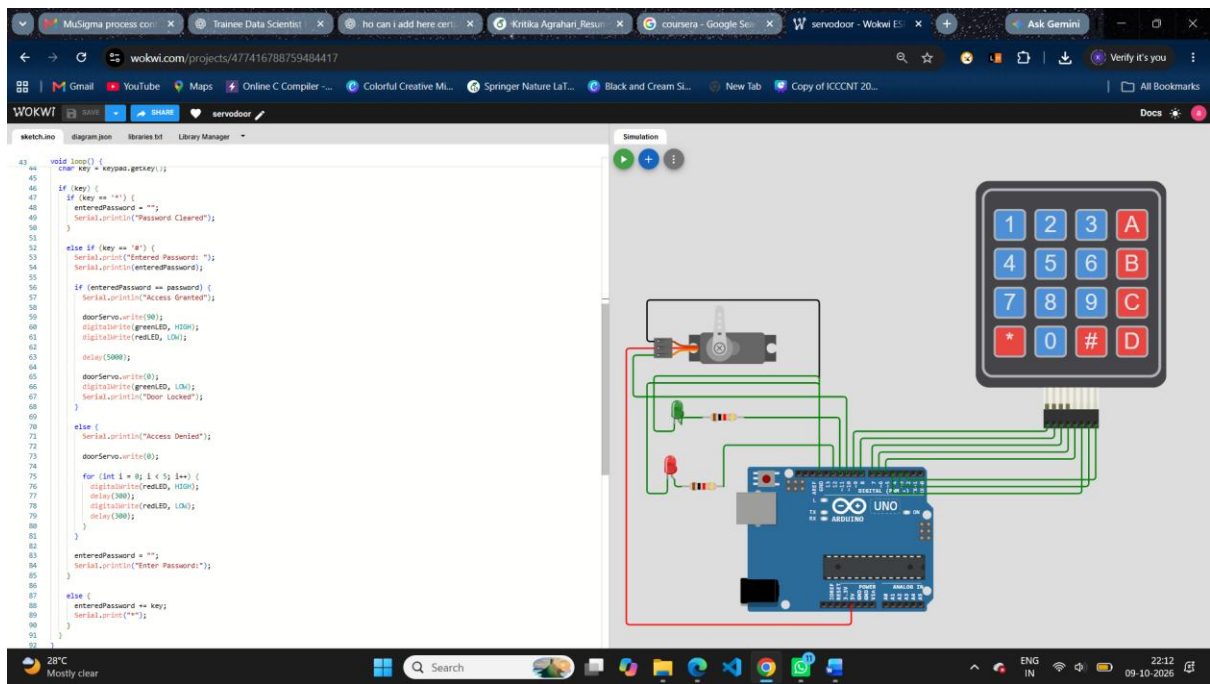
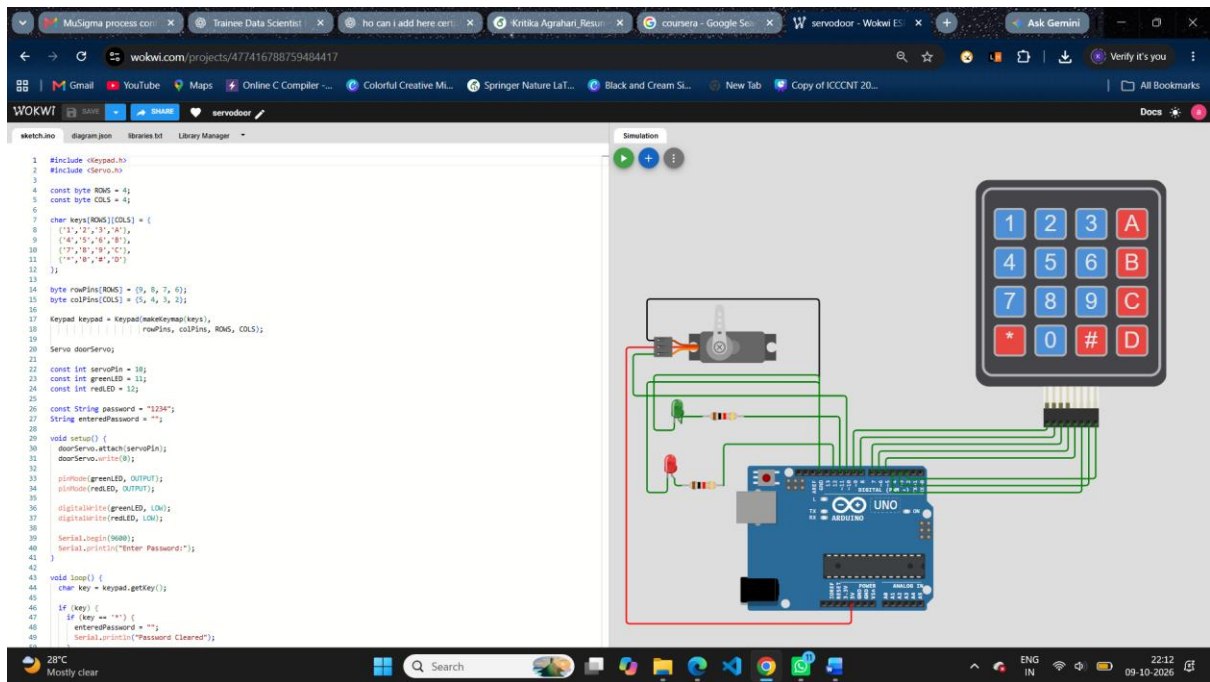








Wokwi IDE interface showing a C++ sketch for an Arduino Uno R3 board. The sketch implements a light sensor (LDR) and a motion sensor (MPJ-01) to control an LED.

Sketch Code:

```
1 const int ldrPin = A0;
2 const int ledPin = 2;
3 const int ledPin = 13;
4
5 const int lightThreshold = 500;
6
7 void setup() {
8   pinMode(ldrPin, INPUT);
9   pinMode(ledPin, OUTPUT);
10  pinMode(ledPin, OUTPUT);
11  digitalWrite(ledPin, LOW);
12  Serial.begin(9600);
13 }
14
15 void loop() {
16   int lightValue = analogRead(ldrPin);
17   int motion = digitalRead(mprPin);
18   Serial.println("LDR Value: ");
19   Serial.println(lightValue);
20   Serial.println("Motion: ");
21   Serial.println(motion);
22
23   if (motion == HIGH) {
24     Serial.println("Detected");
25   } else {
26     Serial.println("No Motion");
27   }
28
29   // Dark = Motion = LED ON
30   if (lightValue < lightThreshold && motion == HIGH) {
31     digitalWrite(ledPin, HIGH);
32   }
33   else {
34     digitalWrite(ledPin, LOW);
35   }
36   delay(500);
37 }
```

Simulation Diagram:

The diagram shows an Arduino Uno R3 board connected to an LDR sensor module and an MPJ-01 motion sensor module. The LDR sensor's VCC is connected to the 5V pin, and its GND is connected to the GND pin. The MPJ-01 motion sensor's VCC is connected to the 5V pin, and its GND is connected to the GND pin. The LDR sensor's output is connected to the A0 pin. The MPJ-01 motion sensor's output is connected to the 2 pin. The LED is connected to the 2 pin (anode) and the 13 pin (cathode).